

# ERIC LU

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## Experience

### Adobe

November 2025 – Present

*Software Engineer: Risk Platform*

*San Francisco, CA*

- Revamped Adobe's Risk Platform front-end SDK: enabled collection of user and browser signals, bot detection, IP intelligence, and fraud vendor integrations across 100M+ \*.adobe.com sessions weekly
- Automated fraudulent email classification pipeline, expanded detection coverage across 4M+ domains, (\$XXM ARR)
- Built Gardener: scheduled agents for continuous codebase health improvements. Fixed security vulnerabilities, memory leaks, thread safety issues across many codebases in Adobe ecosystem
- Developed back-end fraud vendor integrations and risk decisioning: design decisions for parallelism, resiliency, caching, etc. Directly resulted in 400+% fraudulent payment reduction, (\$XXM ARR)

### Adobe

August 2024 – November 2025

*Software Engineer: Payments & Order Processing Services*

*San Jose, CA*

- Designed and built eCommerce platform features: Pause/Resume contract scheduling (\$XXM ARR), daily/weekly App Store subscriptions (\$XXM ARR), undo cancellation (\$XXM ARR), free offer promotion bundling (\$XM ARR)
- Built 1st-place org-wide hackathon-winning Commerce AI agent: enabled E2E checkout, product discovery, upsell, and PayPal-integration through a natural language chat interface
- Led migration to Ethos Flex: CI/CD pipeline with infrastructure-as-code, progressive canary rollouts, automated rollback, and built-in security guardrails. Deployment velocity increased  $\geq 2x$  while maintaining 4 9's availability
- Led Commerce MCP project: Unified interface for AI-assisted support, log analysis, CSO remediation, and observability

### Adobe

May 2023 – August 2023

*Software Engineer Intern: Payments*

*San Jose, CA*

- Designed and tested multiple ML classification models (Neural nets, XGBoost) using Spark Tensorflow and Elephas to drive retry cost reduction and reduce model latency by 50% using distribution at scale through PySpark RDDs
- Built end-to-end data mining processes on payment data to optimize retry schedule with 42 recommendations

### Valley Consulting Group (UC Berkeley)

2021 – 2023

*Project Lead / Consultant*

*Berkeley, CA*

- **Starbucks (Emerging Technologies)**: Blockchain and Web3 consulting for Starbucks Odyssey
- **Roku (Voice)**: ML customer segmentation (k-means, PCA, decision trees), product penetration for 1M+ Roku users
- **Airbnb (Privacy)**: scraped competitor platforms, UI/UX research, increased accessibility of user data management

### U.S. Venture (Breakthrough)

May 2022 – August 2022

*Data Engineer Intern*

*Green Bay, WI*

- Deployed an end-to-end geospatial index for pricing methodology team by integrating common routing REST API, enabling internal query operations to a NoSQL (GeoJSON) internal database built on an R-Tree data structure
- Implemented ETL pipeline processes through Python in Google Cloud Platform, reducing pricing latency by 40%

## Education

### University of California, Berkeley

*B.A. Computer Science, B.A. Data Science, Concentration in Business/Industrial Analytics*

*GPA: 3.7*

Database Systems · Computer Security · Algorithms Analysis · Operating Systems · Machine Learning · Discrete Math · Data Structures · Data Science Principles · Artificial Intelligence · Natural Language Processing · Information Systems

## Projects

**Tow-Away Alerts** | *TypeScript, Firebase Functions, Firestore, Node.js*

2026

- Built an SF parking alert service: daily scheduled job polls city permit data (DataSF sftu-nd43), normalizes and upserts records into Firestore, and triggers email alerts when new tow-away zones are posted near a user's parked car location
- Designed a geospatial matching engine using geohash indexing for curb-segment proximity lookups, with confidence scoring (same block face vs. nearby corridor) and deduplication to prevent repeat alerts for the same permit

## Technical Skills and Interests

**Languages:** Python, Java, JavaScript/TypeScript, C, HTML/CSS, SQL, GraphQL

**Frameworks/Tools:** Spring Boot, Node.js, Flask, React, Apache Kafka, Google ADK, LangChain, MCP, RAG, A2A, TensorFlow, NumPy, Pandas, Scikit-learn, Lambda, S3, DynamoDB, SQS, CloudFront, Kubernetes, ArgoCD, Argo Workflows, Docker, Jenkins, Prometheus, Grafana, Splunk, NewRelic

**Interests:** Baseball, Surfing, Poker, Skiing and Snowboarding, Golf, Behavioral Economic Podcasts, Homemade Cooking